

RABIA ZULFIQAR

Computer Science Undergraduate | FAST National University | CGPA: 3.91/4.0

+92-331-6062436

rzulfiqar889@gmail.com

[Rabia Zulfiqar | LinkedIn](#)

[rabiazulfiqar1](#)

EDUCATION

FAST National University | 2023-2027 (CGPA: 3.91/4.0)

Bachelor of Science in Computer Science

PROJECTS

SparkSpace - Multimodal AI Study Platform | FastAPI, React, PostgreSQL, Redis, Docker

- Architected scalable backend with PostgreSQL & Redis caching, reducing query response time by 40%
- Deployed full-stack containerized app (Docker) with real-time data flow via WebSockets
- Built RAG-based AI assistant (85% accuracy) for intelligent document querying.

IntelliLearn - Personalized Learning Platform | Python, Supabase, pgVector Ext, FastAPI

- Built ETL data pipelines handling 500+ files for automated reporting, applicable to banking data ingestion
- Developed recommendation system using pgVector with advanced PostgreSQL extensions & database architecture
- Implemented multimedia processing (summarization & transcription) with scalable FastAPI backend

Interactive Data Analytics Dashboard | Python, Streamlit, Pandas, Plotly, SQL

- Developed multi-page Streamlit EDA web application on 5,000+ records
- Implemented comprehensive data cleaning pipeline handling outliers using IQR, Z-score, and Winsorization methods.
- Designed interactive visualizations with dynamic filtering; applied Spearman correlation for automated insights

EXPERIENCE

Research Assistant - ComputerVision | FAST National University | Feb 2026 - Present

- Investigating lightweight AI architectures & optimization strategies for efficient on-device image processing
- Conducting literature reviews & experimental analysis, building research, documentation & problem-solving skills

Team Lead - NLP Division | ACM AI Society , FAST University | 2025 - Present

- Led planning & execution of AI seminars and workshops, coordinating cross-functional teams & communications
- Managed timelines and deliverables for multiple concurrent events, demonstrating strong project management
- Developed leadership and teamwork skills in a technical student organization

SKILLS

- | | |
|-----------------------------|--|
| • Data Analytics | SQL, Excel, Pandas, Plotly, Scipy, Data Visualization, Statistical Analysis, ETL pipelines |
| • Databases | PostgreSQL, MySQL, MongoDB, Redis, Database Design & Architecture, Supabase |
| • Programming | Python, Javascript, C, C++ |
| • Backend & APIs | FastAPI, Flask, RESTful APIs, Docker |
| • Frontend | React, Tailwind CSS |
| • IT & Systems | Git/Github, Linux, System Design, OOP, Data Structures & Algorithms |

Soft Skills Team Leadership, Project Management, Communication, Problem-Solving, Documentation

CERTIFICATES

Machine Learning Specialization (3 courses)

Generative AI